

Curriculum Vitae

Jonathan Kingslake

108D Oceanography, Lamont-Doherty Earth Observatory, Palisades, NY

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Education

PhD in Glaciology

Department of Geography, University of Sheffield, UK

Sep 2009–Jan 2013

PhD title: “Modelling Ice-dammed Lake Drainage” with Felix Ng and Grant Bigg.

Undergraduate Physics Degree

Department of Physics, University of York, UK

Sep 2006–May 2009

Grade: First-class with Distinction.

Employment

Associate Professor in Glaciology

Department of Earth Science and Engineering, Imperial College London, UK

Starting in June 2026

Associate Professor with tenure

Department of Earth and Environmental Sciences, Columbia University, USA

July 2023–May 2026

Associate Professor without tenure

Department of Earth and Environmental Sciences, Columbia University, USA

May 2021–June 2023

Assistant Professor

Department of Earth and Environmental Sciences, Columbia University, USA

Jan 2016–April 2021

Glacier Geophysicist

British Antarctic Survey, Natural Environmental Research Council, UK

Jan 2013–Dec 2015

Professional Activities

Recent prizes

- Lenfest Distinguished Faculty Award, Columbia University, 2023

A prize which “recognizes the excellence of Arts and Sciences faculty as teachers, scholars, and mentors within and outside the classroom” and comes with a \$75,000 stipend.

- Division of Natural Sciences Award for Commitment to Service, Columbia University, 2024

A prize that honors “work and dedication to your students, colleagues, and department” and comes with a \$500 unrestricted research fund.

Reviewer

- Nature
- Science Advances
- Nature Climate Change
- US National Science Foundation
- Antarctica New Zealand
- UK Natural Environment Research Council
- The Royal Society
- Geophysical Research Letters
- Philosophical Transactions, A.
- Journal of Geophysical Research
- Journal of Glaciology
- Annals of Glaciology
- International Journal of Applied Earth Observations and Geoinformation
- The Cryosphere
- Journal of Advances in Modeling Earth Systems

Editorial board memberships

- Journal of Glaciology, Jan 2019-2022
- Annals of Glaciology, special issue on Radioglaciology, 2020

Panel member

- US National Science Foundation
- US National Aeronautics and Space Administration

Workshop organization

- NSF-funded workshop on Antarctic Surface hydrology, Columbia, Feb 2018
- *Modelling In The Polar Sciences* workshop, Sheffield, Apr 2012

Sessions co-convened

- American Geophysical Union Fall meeting Dec 2016: Hydrology of Glaciers and Ice Sheets
- American Geophysical Union Fall meeting Dec 2018: Beyond Ice Thickness: Using Radar Sounding to Understand the Dynamics of Glacier Systems
- European Geoscience Union General Assembly Apr 2025 (upcoming): Boundary Conditions and Cryospheric Processes: Investigating Earth-Ice Interactions Beneath Ice Sheets, Shelves and Glaciers

Invited seminars

- NASA Goddard Space Flight Center, Apr 2026: West Antarctic Ice Sheet readvance in the Holocene
- New York University, Mar 2026: West Antarctic Ice Sheet readvance in the Holocene
- University of South Florida, Jan 2026: West Antarctic Ice Sheet readvance in the Holocene
- Massachusetts Institute of Technology, Sep 2025: West Antarctic Ice Sheet readvance in the Holocene
- Cambridge University, UK, March 2025: West Antarctic Ice Sheet readvance in the Holocene
- Sheffield University, UK, Feb 2025: West Antarctic Ice Sheet readvance in the Holocene
- Bristol University, UK, Dec 2024: Observing englacial deformation with radar
- Edinburgh University, UK, Nov 2024: West Antarctic Ice Sheet readvance in the Holocene
- Oxford, Feb 2021: Antarctic Surface Hydrology and Its Impact on Ice-Sheet Mass Balance
- MIT, Nov 2020: Antarctic Surface Hydrology and Its Impact on Ice-Sheet Mass Balance
- ETH, June 2020: Antarctic Surface Hydrology and Its Impact on Ice-Sheet Mass Balance
- Princeton University, June 2019: Antarctic Surface Hydrology and Its Impact on Ice-Sheet Mass Balance

- Los Alamos National Lab, June 2019: Antarctic Surface Hydrology and Its Impact on Ice-Sheet Mass Balance
- Poughkeepsie, Marist Center for Lifetime Study, Oct 2017: Present and Past Flow of the Antarctic Ice Sheet
- Edinburgh University, UK, Nov 2015: Ice-Flow Reorganization in the SW Weddell Sea, West Antarctica, 2.5 ka BP
- University of Sheffield, May 2015: Dating Late Holocene Ice-Rise Formation and Grounding Line Retreat in the SW Ronne Ice Shelf Region, Antarctica
- Columbia University, June 2014: Using Modelling, Satellite Data, and Radar to Understand Ice Sheet Processes
- Cambridge, Mar 2014: Radar Investigations of Antarctic Ice Rises
- Rothera Research Station, Antarctica, Jan 2014: Sledge November South Ronne Traverse
- University of Oxford, Nov 2012: Environmental Controls on Jökulhlaups Characteristics
- University of Manchester, Nov 2012: Environmental Controls on Jökulhlaups Characteristics
- Cold and Arid Regions Environmental and Engineering Research Institute, Lanzhou, China, July 2010: Dynamics of Subglacial Outburst Floods

Research Funding

Total as PI: \$8,043,665 (approx. £6,038,017)

Total as co-PI or PI: \$13,425,296 (approx. £10,414,284)

Upcoming

PI: Antarctic surface hydrology and sea-level rise

The Royal Society

Amount: £4,008,479 (approx. \$5,340,016)

Duration: 09/01/2026–08/31/2031

Current

Co-PI: Modeling the coupled dynamics of groundwater, subglacial hydrology, and ice sheets

NSF

Amount: \$356,512 (approx. £282,452)

Duration: 09/01/2024–08/31/2026

PI: S. Larochelle (UCLA)

My role: assisted postdoc Stacy Larochelle with preparation of the proposal and working on the project in a mentoring role.

PI: NSFGE0-NERC: Investigating the direct influence of meltwater on Antarctic Ice Sheet dynamics

NSF

Amount: \$899,856 (approx. £712,937)

Duration: 08/15/2023–07/31/2026

Co-PIs: A. Sole (University of Sheffield), S. Livingstone (University of Sheffield), J. Ely (University of Sheffield), K. Winter (Northumbria University)

My role: led preparation of the proposal and leading the project.

PI: STC: Learning the Earth with Artificial Intelligence and Physics (LEAP)

NSF

Amount: \$113,844 (approx. £90,194)

Duration: 04/01/2022–09/01/2025 (recently extended)

No co-PIs

My role: led preparation of the proposal and leading the project.

PI: COMBINED: Channelized Ocean Melting Beneath Ice Shelves: Non-hydrostatic Ice and Estimating Shelf Densities

NASA

Amount: \$603,355 (approx. £478,018)

Duration: 01/01/2022–12/31/2025

Co-PIs: K. Tinto (Columbia), R. Constantino (Columbia)

My role: assisted the initial PI, David Porter, with conceiving and preparing the proposal. Now that Porter is working at NSF, I am leading this proposal.

Co-PI: Collaborative Research: Sensitivity of the West Antarctic Ice Sheet to 2°C (SWAIS 2°C)

NSF

Lamont budget: \$491,568 (approx. £389,453)

Duration: 09/15/2021– 08/31/2026

PI of Columbia University sub-award: J. Austermann (Columbia)

Lead PI: M. Patterson (Binghamton)

My role: assisted the PIs in the preparation of the proposal and now responsible for geophysics on the Crary Ice Rise, Antarctica.

Co-PI: Understanding firn rheology through laboratory compaction experiments and radar data

NSF

Amount: \$737,532 (approx. £584,322)

Duration: 01/01/2020–03/21/2025

PI: C. McCarthy (Columbia)

My role: assisted the initial PI, Rob Skarbek, with conceiving and preparing the proposal.

Co-PI: Supplement to Understanding firn rheology through laboratory compaction experiments and radar data

NSF

Amount: \$87,301 (approx. £69,156)

PI: C. McCarthy (Columbia)

My role: assisted the current PI, Christine McCarthy, with preparing the proposal to fund a postbaccalaureate student project.

PI: Accelerating data science approaches in glaciology

Columbia's Earth Institute and The Center for Climate and Life

Amount: \$135,622 (approx. £107,448)

Awarded in 2020 (ongoing)

No co-PIs

My role: led preparation of the proposal and leading the project.

Co-PI: NNA Track 1: Predicting coastal response to a changing Greenland Ice Sheet

NSF

Amount: \$2,986,095 (approx. £2,365,785)

Duration: 09/01/2019–08/31/2024

PI: K. Tinto (Columbia)

Co-PIs: J. Austermann (Columbia), R. Bell (Columbia)

My role: assisted the PIs in the preparation of the proposal.

PI of Columbia University sub-award: Ground Geophysics Survey of Thwaites Glacier

NSF-NERC

Amount: \$249,535 (approx. £197,698)

Duration: 04/01/2018–08/31/2026

Lead US PI: S. Anandakrishnan (Penn State University)

My role: collaborated with preparation of proposal and leading the portion of the project tasked with collecting and analyzing phase-sensitive radar data.

PI of Columbia University sub-award: Supplement to Ground Geophysics Survey of Thwaites Glacier

NSF

Amount: \$33,856 (approx. £26,823)

Title: Additional ApRES units for GHOST traverse

My role: assisted postdoc Andrew Hoffman and grad-student Elizabeth Case in their first NSF submission.

PI of Columbia University sub-award: Supplement to Ground Geophysics Survey of Thwaites Glacier

NSF

Amount: \$76,630 (approx. £60,711)

Title: Glacial Hauntologies: asymmetrical echoes of ice across time and space

My role: assisted postdoc Andrew Hoffman and grad-student Elizabeth Case with the submission.

Past

Co-PI: Greenland Ice Sheet dynamic response to inland expansion of a hydrologically-active ice-sheet bed

NSF

Amount: \$722,623 (approx. £572,433)

Duration: 09/01/2020–09/30/2025

PI: M. Nettles (Columbia)

My role: assisted the initial PI, Laura Stevens, with conceiving and preparing the proposal.

PI: Satellite observations and modelling of surface meltwater flow and its impact on ice shelves

NSF

Amount: \$535,885 (approx. £424,506)

Duration: 08/01/2018–07/31/2021

No co-PIs

PI: Measuring snow compaction to better quantify ice-sheet changes

Columbia, Lenfest junior faculty development grant

Amount: \$14,990 (approx. £11,874)

Duration: 01/01/2018–12/31/2018

No co-PIs

My role: led preparation of the proposal and leading the project.

PI: Workshop on Antarctic Surface Hydrology and Future Ice-Shelf Stability

NSF

Amount: \$40,076 (approx. £31,747)

Duration: 07/01/2017–06/31/2018

Co-PIs: L. Trusel (Penn State), A. Banwell (University of Colorado Boulder), R. Bell (Columbia), I. Das (Columbia), R. DeConto (UMass Amherst), M. Tedesco (Columbia), J. Lenaerts (University of Colorado Boulder), C. Schoof (University of British Columbia)

My role: led preparation of the proposal and leading the project.

Pending

Co-PI: Novel processing of Operation IceBridge radar data to constrain the deformation, structure, and genesis of basal units in Antarctica and Greenland

NASA

Amount: \$454,388 (approx. £359,954) (3 years)

PI: A. Hoffman (Columbia/Rice)

My role: assisted postdoc Andrew Hoffman with preparation of the proposal and will work on the project in a mentoring role if it is funded.

Co-PI: Collaborative Research: Constraining ice rheology and basal resistance using englacial vertical velocity observations

NSF

Amount: \$90,425 (approx. £67,726) (3 years)

PI: A. Hoffman (Columbia/Rice)

My role: assisted postdoc Andrew Hoffman with preparation of the proposal and will work on the project in a mentoring role if it is funded.

Project partner: Open-source Pipeline for ApRES Archiving and Processing

Astera institute

Amount: \$99,593 (approx. £74,100) (1 year)

My role: assisted postdoc Elizabeth Case in formulating the idea and will help with code development if it funded.

Teaching

Columbia University

(Where available, the class size and overall student assessment out of five is provided below. Quotes are from student evaluations with a prompt starting with “Please evaluate Jonathan Kingslake...”)

Introduction to Solid Earth System Science (EESC4220)

This class is a co-taught (between two faculty members and several TAs), long-running undergraduate lecture and lab class, with a one-day field class in New York City and surroundings. I took over from another faculty member in 2016 and adapted and expanded the class over 8 years.

(2016-2023):

- 2016, ~40 students
- 2017, 36 students
- 2018, 37 students
- 2019, 36 students, 3.88/5
 - “Honestly, Prof. Kingslake was my best professor this semester.”
- 2020 (taught virtually), 53 students, 4.58/5
 - “Professor Kingslake's lectures were always incredibly interesting and clear”
 - “He is SO kind and helpful, especially in office hours.”
- 2021, 48 students, 3.97/5
 - “I found his lectures incredibly engaging.”
- 2022, 46 students, 4.52/5
 - “Professor Kingslake is a truly excellent professor.”
- 2023, 70 students, 4.19/5
 - “Jonny is THE BEST! He brings so much energy into his lectures and makes sure that we really understand the material.”
- 2025, 34 students, 4.41/5

Glaciology (EESC4220)

This class is half lectures and half computer practicals.

I designed and taught this class for 5 years between 2018 and 2023, not in 2022 due to parental leave:

- 2018, 8 students
 - 2019, 7 students, 4.75/5
 - “Jonny's passion for glaciology clearly shines through his lectures.”
 - “One thing that is fantastic that we do in class is with going over derivations of the equations that describe glacial behavior.”
 - 2020 (taught virtually), 6 students
 - 2021 (taught virtually), 6 students, 5/5
 - “Jonny is a great professor [...] I think he did a great job lecturing, despite it being totally over zoom!”
 - 2023, 14 students, 4.89/5
 - “Jonny is an incredibly effective lecturer, and it's apparent how much he cares about both the material and his students”
 - 2025, 11 students, 5/5
- (Code and slides can be found [here](#).)

Glacial Processes Seminar (EESC9815)

This was a class based around discussion and paper-reading that I co-taught with another faculty member.

- 2017, ~8 students and postdocs

Theory of Ice Sheet Dynamics (EESC9920)

This was a class based around lectures and discussion that I developed as an advanced version of my main glaciology class (*EESC4220*) with another faculty member.

- 2022, 6 students (+several postdocs), 4.67/5
 - (Notes and other material can be found [here](#))

- 2023, 3 students (+several postdocs)

Death Valley Field Trip (UV1010)

Undergraduate field class, Mar 2017

Field Geology (GR8010)

Postgraduate field class to Peru, July 2019

Guest lectures

- MPA Environmental Science and Policy, 2018
- Columbia Climate School, Polar Warming and Ice Sheet Melt, Spring 2022
- EESD Sustainable Development, 2016-2024

Sheffield University

- *Environmental Modelling* (Lecturer)
- *Atmospheres and Oceans* (Field Demonstrator)
- *Dryland Geomorphology* (Practical and Computer Lab Demonstrator)
- *Practical Methods for Physical Geographers* (Field Demonstrator)
- *Research Design in Physical Geography* (Tutor)
- *Environmental Change* (Tutor)

Student supervision

I have been lead-advisor to three graduate students and on the mentoring committees of nine others. I have also mentored seven undergraduate students and one masters student.

Graduate students

George Lu

- PhD
- Fall 2021 – present
- Title: Modeling and Observations of Ice Flow
- Topics: surface lakes in Greenland and subglacial hydrology, subglacial hydrology and ice dynamics, ice rheology and physics-informed neural networks
- I am George's primary advisor
- Co-advised by M. Nettles (Columbia)
- Funding: NSF and NSERC graduate research fellowship

Julian Spergel

- PhD
- Fall 2016 – spring 2022
- Title: Modelling and remote sensing of meltwater drainage on Antarctic ice shelves
- I was Julian's primary advisor
- Funding: NSF
- Now a hydrologist at the Georgia Environment Agency, USA

Elizabeth Case

- PhD
- Fall 2017 – spring 2024
- Title: Ice: formation, deformation, and disappearance
- Topics: firn compaction, ice-penetrating radar, Antarctic ice streams, ice fabric, US mountain glaciers

- I was Elizabeth's primary advisor
- Funding: NSF graduate research fellowship

Gabriel Traietti

- PhD
- Fall 2025 – present
- Topics: Southern Ocean paleoclimate
- I am one of *Gabriel's* co-advisors
- Primary advisor: G. Winkler (Columbia)

Casey Brayton

- PhD
- Fall 2021 – present
- Title: Community-Centered Methods of Understanding Local and Regional Sea Level Change for Data-Driven Adaptation in the Arctic
- Topics: bathymetry, co-production, tides, Greenland communities
- I am one of Casey's co-advisors
- Primary advisor: K. Tinto (Columbia)
- Funding: NSF

Caitlin Locke

- PhD
- Fall 2021 – present
- Title: TBD
- Topics: airborne geophysics, sub-ice-shelf bathymetry
- I am one of Caitlin's co-advisors
- Primary advisor: K. Tinto (Columbia)
- Funding: NSF

Charlotte Bate

- PhD
- Fall 2023 – present
- Title: TBD
- Topics: ice-friction experiments
- I am one of Charlotte's co-advisors
- Primary advisor: C. McCarthy (Columbia)

Emily Glazer

- PhD
- Fall 2021 – present
- Title: An Examination of Ice-Shelf Stability through Remote Sensing and Numerical Modeling
- Topics: ice-shelf dynamics, flexure, numerical modelling, altimetry, surface melting
- I am one of Emily's co-advisors
- Primary advisor: R. Buck (Columbia)
- Funding: NASA

Aaron Stubblefield

- PhD
- Fall 2017 – 2022
- Title: Modelling the dynamics and surface expressions of subglacial water flow
- Topics: numerical modelling, subglacial hydrology, ice dynamics

- I was one of Aaron's co-advisors
- Primary advisor: T. Creyts (Columbia)
- Funding: NSF, NASA

Roger Creel

- PhD
- Fall 2018 – fall 2023
- Title: Reconstructing and understanding how past warming affected sea level, ice sheets, and permafrost
- Topics: GIA, sea level, Bayesian statistics, ice-sheet and sea-level history
- I was one of Roger's co-advisors
- Primary advisor: J. Austermann (Columbia)

Alexandra Balter

- PhD
- Fall 2018 – spring 2023
- Title: Evaluation of ice sheet vulnerability and landscape evolution using novel cosmogenic-nuclide techniques
- Topics: exposure dating, history of Greenland
- I was one of Allie's co-advisors
- Primary advisor: J. Schaefer (Columbia)

Theresa Sawi

- PhD
- Fall 2018 – fall 2023
- Title: Unsupervised Machine-Learning Applications in Seismology
- Topics: machine learning, seismology, subglacial hydrology, earthquakes
- I was one of Theresa's co-advisors
- Primary advisor: F. Waldhauser (Columbia)

Masters and undergraduate students

Hengbo Tong

- Masters research opportunity
- Spring-fall 2023
- Topic: ice-sheet basal processes and machine learning

Dean Gelling

- Undergraduate final thesis project
- Fall 2023 – spring 2024
- Topic: mountain glacier evolution in high-mountain Asia

Chiemi Tagami

- Undergraduate final thesis project
- Fall 2022 – spring 2024
- Topic: readvance of the West Antarctic Ice Sheet

Janette Levin

- Undergraduate research opportunity
- Jan 2019-May 2021
- Topic: measuring and modelling the shape of Antarctic ice ridges

Eileen Chen

- Undergraduate research opportunity

- Fall 2017
- Topic: using fluid-mechanical experiments to gain insight into glaciological processes

Peter Richards

- Undergraduate research opportunity
- Fall 2017
- Topic: augmented reality in Earth Science

Liang Fu

- Undergraduate research opportunity
- Fall 2016
- Topic: constraining the ice flow and climate history of the Antarctic Ice Sheets using ice-penetrating radar

Eleni Reynolds

- Undergraduate research opportunity
- Fall 2016
- Topic: constraining the ice flow and climate history of the Antarctic Ice Sheets using ice-penetrating radar

Public outreach

- Started an open-source, online textbook called the [Glaciology Data Analysis and Modeling book](#), 2022–present
- Maintaining a website showcasing Antarctic fieldwork on Flask Glacier, Sep 2023–present
- Seminar, City Tech High School Summer program, Brooklyn, New York, *Ice Flows*, July 2023
- Lamont Open House, 2016-2023
- Science advisor on a recently filmed glacier-themed short movie by Stephanie Falkeis, Columbia MFA Film alum, 2020–2022
- Panel following public screening of *Antarctica: Ice and Sky*, People and the Planet in Francophone Cinema, Oct 2022
- Virtual seminar, EI-Live, *Ice flows: Using math and physics to understand how*, Nov 2021 ([YouTube link](#))
- Live demo innovative teaching tool: Lamont’s lightboard presentation studio, *Ice-shelf Melting*, June 2021 ([video link](#))
- Virtual panel on fieldwork food, EI-Live, April 2021
- Virtual seminar, EI-Live, *The Earth’s Blanket — Physics of Glaciers*, Mar 2021 ([YouTube link](#))
- Virtual seminar, Charter School event, Ice Sheets, Oct 2020
- Virtual seminar, EI teach series (High School educator training), *Shrinking Glaciers, Growing Science*, Aug 2020
- Virtual seminar, Columbia Alumni Association’s Columbia@home series, July 2020
- Virtual seminar, EI-Live, *Past and present changes in Antarctica*, April 2020
- Virtual seminar, EI-Live, *Virtual field trip to Juneau Alaska*, April 2020
- Virtual seminar for the NYC Department of Education’s Keepin’ Sustainability Alive Inside series, April 2020
- Lecture, Horace Mann High School, Jan 2020
- Public lecture, Bruce Museum, Connecticut, Oct 2019
- Antarctica week, school talk at Middle Creek Elementary School and Cottage Lane Elementary School, Dec 2018, Dec 2019
- Earthfest, American Museum of Natural History, April 2019
- Science advisor on an Antarctic-themed screenplay by Lauren Fondren, Columbia MFA Film student, 2018–2019
- Open House educator event, Oct 2019
- Women in STEM at the Intrepid Air and Space Museum, Mar 2016, Mar 2018

- Career day at the Lycée Français School, New York, May 2016, May 2017
- Visit to a summer school for high-school students at the Hudson River Community Sailing Club, May 2016
- Earth-Sun Day at the American Museum of Natural History, Mar 2016
- Designed and ran Antarctica-themed activity days for 50+ students at schools around the UK, 2013–2015

Media

- According to metrics compiled by *Nature*, our paper *Widespread movement of meltwater onto and across Antarctic ice shelves* was featured in 92 news stories across 70 outlets including The Washington Post, Wall Street Journal, National Geographic, The Atlantic magazine, and The Independent (UK) (April 2017).
- Another recent paper *Vulnerability of Antarctica's ice shelves to meltwater-driven fracture*, also in *Nature*, was featured by 51 news outlets, including in the New York Times, Scientific American, New Scientist, and The Guardian.
- Other media appearances include Fox 5 News (on-screen interview, July 2017), National Public Radio (I recorded a short commentary on my work that was broadcast in Sep 2017), and SiriusXM (live interview Sep, 2020).
- Scientific American produced a podcast about our Alaska fieldwork in July 2018 ([link](#)).

Administrative and Community Service

- Chair of Diversity, Equity, and Inclusion committee, Department of Earth and Environmental Sciences, 2023-2024
- Columbia graduate student (non-mentees) exam committees:
 - Christopher Carchedi, oral qualifying exam committee, July 2020
 - Max Cunningham, defense committee, June 2019
 - Kira Olsen, defense committee, Oct 2019
 - Joshua Maurer, oral qualifying exam committee, Feb 2017
- Lamont Diversity, Equity and Inclusion Task force, July 2020–Dec 2020
- Lamont vision committee, March 2020–Nov 2020
- Promotions and Careers committee (member) Nov 2019–Jan 2022
- Graduate student admissions committee, Department of Earth and Environmental Sciences, 2017–2021
- Strategic planning committee, Department of Earth and Environmental Sciences, 2018
- Chair of search committee for postdoctoral research scientist, Nov 2020–Nov 2021
- Chair of search committee for postdoctoral research scientist, fall 2023
- Chair of search committee for postdoctoral research scientist, spring 2024
- Search committee for Associate Research Scientist, NASA GISS, chair: G. Schmidt, July 2020
- Search committee for postdoctoral research scientist, chair: J. Austermann, July 2019
- Search committee for postdoctoral research scientist, chair: T. Creyts, Nov 2018
- Search committee for postdoctoral research scientist, chair: K. Tinto, Oct 2018
- Search committee for postdoctoral research scientist, chair: M. Tedesco, Oct 2017
- Search committee for faculty hire in experimental Earth science, Department of Earth and Environmental Sciences, chair M. Spiegelman, Jan 2017
- Chair of search committee for postdoctoral research scientist, Aug 2016
- Member of the International Thwaites Glacier Collaboration's committee on Inclusion, Diversity, Equity, and Access, 2020–2023.
- External examiner: Cornelius Kaspar, Masters thesis defense, July 2025
- External examiner: Rohi Muthyala, PhD defense, Rutgers University, April 2023
- External examiner: David Bigelow, Masters examination, Simon Fraser University, Mar 2019

Fieldwork

Alaska

- July 2018
- Ice-penetrating radar
- Ice coring
- Measuring snow density and densification

Greenland

- April–May 2017
- Ice-penetrating radar surveys
- Drilling 20–30 m ice cores
- Installed weather stations

Antarctica

- Nov 2013–Jan 2014 and Nov 2014–Jan 2015
- Planned and led a two-person, 2500-km skidoo traverse of the Ronne Ice Shelf
- Used radar to measure the internal stratigraphy and flow of the ice sheet

Southern Norway

- July 2011
- Investigated the origin of ogive bands on Jostedalsbreen with structural and photographic surveys

Svalbard

- June 2010
- Hydrological monitoring and sampling
- Surface spectrometry
- GPS surveys

Personal Development and Developing Others

Postdoctoral researcher mentoring

I have mentored 12 postdoctoral researchers (I was/am primary mentor to eight of these 12). Of the eight postdoctoral researchers who have left Lamont, seven are in faculty positions, including at world-leading universities including Oxford, Stanford, UCL, and one is a program manager at ESA.

Rohi Muthyala

- April 2024–present
- I am Rohi's primary mentor
- Topics: Antarctic ice dynamics, weather observations, uncrewed aerial vehicles in Antarctica
- Funding: NSF-NERC

Racheet Matai

- Dec 2023–present
- I am Racheet's primary mentor
- Topics: ice-sheet modeling, machine learning, basal sliding
- Funding: NSF

Andrew Hoffman

- Apr 2023–present
- I am Andrew's primary mentor
- Topics: ice sheet dynamics, modeling and field observation, ice-shelf basal melting
- Funding: NSF and NASA

Stacy Larochelle

- Sep 2022–June 2025
- I am Stacy's primary mentor
- Topics: Greenland supraglacial lake drainage, subglacial aquifers
- Funding: NSF
- Now on the faculty at UCLA

Sasha Montelli

- Sep 2019–Dec 2020
- I was Sasha's primary mentor
- Topics: thermal evolution of ice rises
- Funding: Schmidt fellowship
- Now on the faculty at University College London, UK

Ching-Yao Lai

- Sep 2018–Oct 2020
- I was Yao's primary mentor
- Topics: fracture mechanics in ice
- Funding: Lamont fellowship and NSF
- Now on the faculty at Stanford University, USA

Martin Wearing

- Jan 2017–Dec 2019
- I was Martin's primary mentor
- Topics: ice dynamics/glacial geophysics
- Funding: institutional funds and NSF
- Now a Digital Twin Earth Scientist at the European Space Agency, Italy

Laura Stevens

- Jan 2018–July 2020
- I was on Laura's mentoring committee
- Topics: dynamics of Greenland outlet glaciers
- Funding: Lamont fellowship
- Now on the faculty at the University of Oxford, UK

Fiona Clerc

- 2022–2024
- I was on Fiona's mentoring committee
- Topics: modelling of ice sheets and the lithosphere
- Funding: Lamont fellowship
- Now a postdoc at Stanford University, USA

Benjamin Keisling

- 2020–2022
- I was on Benjamin's mentoring committee
- Topics: paleo ice-sheet modelling
- Funding: Lamont fellowship
- Now on the faculty at the University of Texas, USA

Guy Paxman

- April 2019–2022
- I was on Guy's mentoring committee
- Topics: paleo glacial landscapes

- Funding: Lamont fellowship
- Now on the faculty at Durham University

Kerry Callaghan

- 2021–2022
- I was on Kerry's mentoring committee
- Topics: modelling the global water table
- Funding: NSF
- Now on the faculty at the University of Illinois Chicago

Career breaks

I took paternal leave for one semester in spring 2022 and spring 2024.

Career development

I continually try to expand my technical skillset. For example, since 2020 I have moved from proprietary software like MATLAB to the open-source Python ecosystem. The fact that Python is open source facilitates reproducible research by allowing anyone with a computer to replicate my work and teaching materials. Through this transition, I also increased my competencies in version control (e.g., Git), which helps maintain organized and transparent code, unit testing, to ensure code reliability and accuracy, and cloud computing, which aid in large-scale data analysis.

This effort has significantly enhanced my scientific and pedagogical capabilities. Specifically, I started a new python-based, radar processing package called [XApRES](#), which along with other advantages facilitates cloud-based analysis of large datasets collected with a radar system called ApRES and is being used by grad-student George Lu. I also started a new online textbook called the [Glaciology Data Analysis and Modeling book](#), which includes text, equations, plot and crucially all the code needed to perform the various data analysis and modelling tasks described. Both the online textbook and XApRES are fully open and can be accessed and contributed to by anyone.

Publications

Refereed full papers

Total: 42

(first author: 9, mentee first author: 17)

Citations: 2657

H-index: 24

(Google Scholar, Mar 26th 2026)

* indicates lead author was a mentee while work was conducted

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